Ch 7.2-7.3: Step Functions and Basis Functions

Lecture 21 - CMSE 381

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Announcements

Last time:

- 7.1 Polynomial regression
- 7.2 Step functions

This lecture:

- 7.2 Step functions
- 7.3 Basis functions

Announcements:

- Vote tomorrow if you're elligible!
- HW # 7 on github

Section 1

Last time

Polynomial regression

$p = 1$

Replace linear model

$$y_i = \beta_0 + \beta_1 x_1 + \varepsilon_i$$

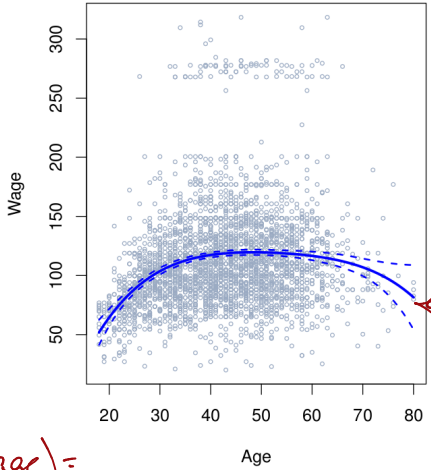
with

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$

$$\begin{bmatrix} | & | & | \\ x & x^2 & x^3 \\ | & | & | \end{bmatrix}$$

$\beta_1 \quad \beta_2 \quad \beta_3$

Example with wage data



f(age)

f(age) =

intercept

-3×10^{-5}

$$-184.1542 + 21.24552 * age + -0.56386 * age^2 + 0.00681 * age^3 + -3e-05 * age^4$$

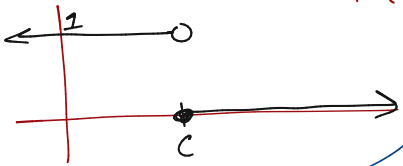
Section 2

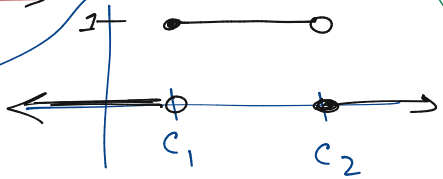
Step Functions

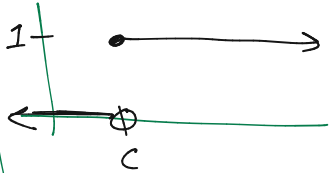
Step functions

$C_0(X) = \mathrm{I}(X < c)$ $\quad$ $\mathrm{I}(c_1 \leq X < c_2)$ $\quad$ $\mathrm{I}(c \leq X)$

$X \in (-\infty, c)$ $\quad$ $X \in [c_1, c_2)$ $\quad$ $X \in [c, \infty)$

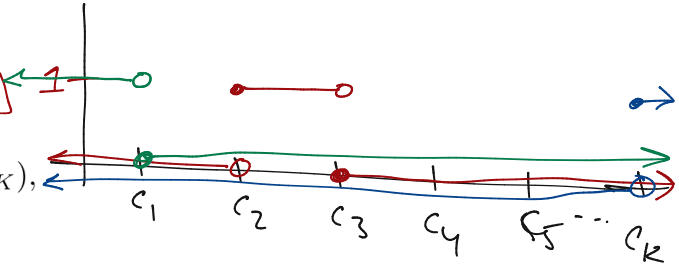






More on step function setup

$$
\begin{aligned}
C_0(X) &= I(X < c_1), \\
C_1(X) &= I(c_1 \le X < c_2), \\
C_2(X) &= I(c_2 \le X < c_3), \\
&\vdots \\
C_{K-1}(X) &= I(c_{K-1} \le X < c_K), \\
C_K(X) &= I(c_K \le X),
\end{aligned}
$$



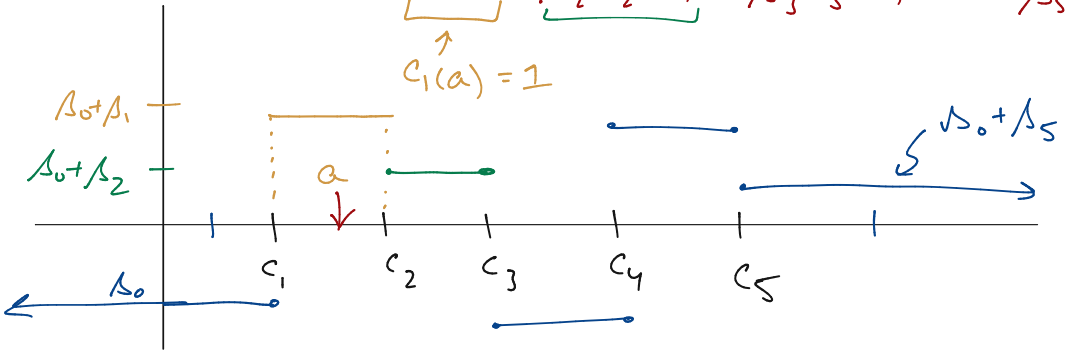
Step function: Learned model

$x \in [c_1, c_2)$ $\quad$ $x \in (c_2, c_3)$ $\quad$ $x \in [c_K, \infty)$

$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i) + \varepsilon_i$$

$$f(x) = \beta_0 + \beta_1 \underbrace{C_1(x)} + \underbrace{\beta_2 C_2(x)} + \beta_3 C_3(x) \ldots + \beta_5 C_5(x)$$

$C_1(a) = 1$



Example

Given knots $c_1 = 3$, $c_2 = 5$, $c_3 = 7$, determine the entries in the columns for $C_i(X)$ in the below matrix.

	$(-\infty, 3)$	$[3,5)$	$[5,7)$	$[7,\infty)$
X	$C_0(X)$	$C_1(X)$	$C_2(X)$	$C_3(X)$
1	1	0	0	0
2	1	0	0	0
3	0	1	0	0
4	0	1	0	0
5	0	0	1	0

	$(-\infty, 3)$	$[3,5)$	$[5,7)$	$[7,\infty)$
X	$C_0(X)$	$C_1(X)$	$C_2(X)$	$C_3(X)$
6	0	0	1	0
7	0	0	0	1
8	0	0	0	1
9	0	0	0	1
10	0	0	0	1

Draw the function

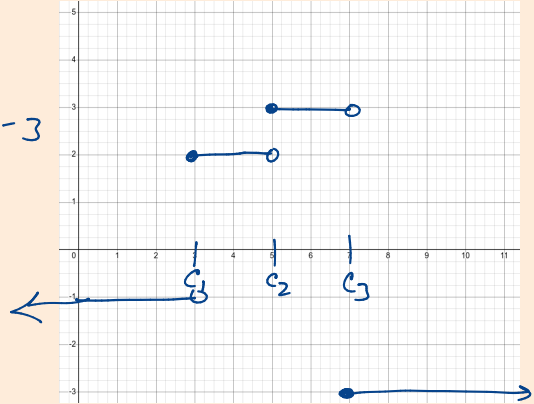
My code doing regression on the step function input returned the function.

$$f(X) = -1 + 3C_1(X) + 4C_2(X) - 2C_3(X).$$

Fill in the table of values, then draw this function below.

X	$f(X)$
1	-1
2	-1
3	$-1+3=2$
4	2
5	$-1+4=3$

X	$f(X)$
6	3
7	$-1-2=-3$
8	-3
9	-3
10	-3



Step function: Learned model

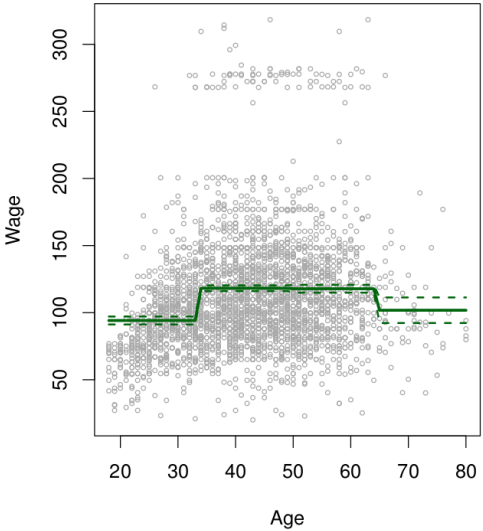
$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i) + \varepsilon_i$$

- We don't use $C_0(x)$ (because we have all other info)
- Code: Not using $C_K(x)$ because our data has $x_i \leq c_K$ (max value) so $C_K(x) = 0$ for all data points

~~Draw the function~~ Coding

Back to the wage data set

Step function example

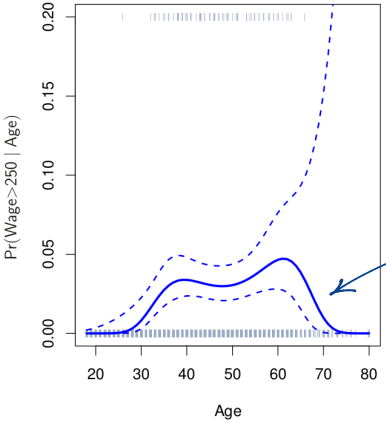


Section 3

Classification versions

Classification version: Polynomial regression

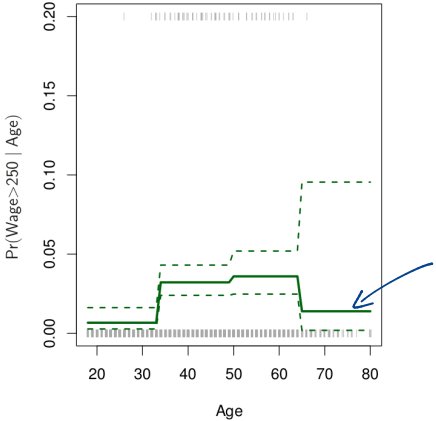
$$\Pr(y_i > 250 \mid x_i) = \frac{\exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}{1 + \exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}$$



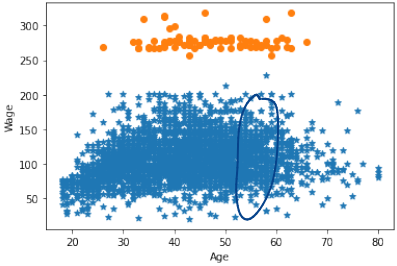
$f(x)$

Classification version: Step functions

$$\Pr(y_i > 250 \mid x_i) = \frac{\exp(\overbrace{\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i)}^{\text{step functions}})}{1 + \exp(\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i))}$$



Computation in the jupyter notebook



In [58]:
```
from sklearn.linear_model import LogisticRegression
```

In [59]:
```
y = np.array(df.wage>250) #<--- this makes sure I
                          #     just have true/false input
clf = LogisticRegression(random_state=48824)
clf.fit(df_steps_dummies,y)
```

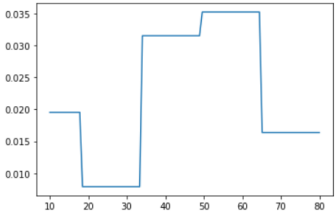
Out[59]:
```
LogisticRegression(random_state=48824)
```

In [56]:
```
f = clf.predict_proba(t_dummies)
plt.plot(t,f[:,1])
```

matrix of dummy variables

```
(100, 2)
```

Out[56]:
```
[<matplotlib.lines.Line2D at 0x7fc2b938bb50>]
```



A few more comments on step functions

- Breaks up the domain so that the functions learned aren't forced to have global structure (good/bad?)
- Make choices about c_i values, how many, where?

Basis Functions Setup

Polynomial and piecewise-constant regression models are special cases of a *basis function* approach.

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_K b_K(x_i) + \varepsilon_i$$

Pick a collection of functions: $\{b_1(x), \ldots, b_K(x)\}$

$\{x\}$ polynomial version

$b_\ell(x) = x^\ell$

Step fcn

$b_\ell(x) = C_\ell(x)$

Next time:

- b_i's are splines
- Wavelets
- Fourier series

Next time

20	F	Nov 4	Polynomial & Step Functions.	7.1,7.2	
21	M	Nov 7	Step Functions	7.2	
22	W	Nov 9	Basis functions, Regression Splines	7.3,7.4	
23	F	Nov 11	Decision Trees	8.1	HW #7 Due
24	M	Nov 14	Ensemble methods	8.2	
25	W	Nov 16	Maximal Margin Classifier	9.1	
26	F	Nov 18	SVC	9.2	HW #8 Due
27	M	Nov 21	SVM	9.3, 9.4, 9.5	
28	W	Nov 23	Single layer NN	10.1	
	F	Nov 25	No class - Thanksgiving		
29	M	Nov 28	Multi Layer NN	10.2	HW #9 Due
30	W	Nov 30	CNN	10.3	
31	F	Dec 2	Unsupervised Learning & Clustering	12.1, 12.4	
32	M	Dec 5	More Clustering	12.4	HW #10 Due
	W	Dec 7	Review		
	F	Dec 9	Midterm #3	Bring your cheat sheet and a non-internet-connected calculator	